

Module-1: Physical Optics: Polarization, Malus' Law, Brewster's Law, Double Refraction, Interference in thin films (Parallel films), Interference in wedge-shaped layers, Newton's rings, Fraunhofer diffraction by single slit, Double slit. Elementary ideas of fibre optics and application of fibre optic cables. [8]

Module-2: Electromagnetic Theory: Gradient, Divergence and Curl, Statement of Gauss theorem & Stokes theorem, Gauss's law, Applications, Concept of electric potential, Relationship between E and V, Polarization of dielectrics, dielectric constant, Gauss's law in magnetostatics, Ampere's circuital law, Equation of continuity, Displacement current, Maxwell's equations. [8]

Module-3: Special Theory of Relativity: Introduction, Inertial frame of reference, Galilean transformations, Postulates, Lorentz transformations and its conclusions, Length contraction, time dilation, velocity addition, Mass change, Einstein's mass energy relation. [6]

Module-4: Quantum Mechanics: Planck's theory of black-body radiation, Compton effect,(qualitative) Wave particle duality, De Broglie waves, Davisson and Germer's experiment, Uncertainty principle, Brief idea of Wave Packet, Wave Function and its physical interpretation, Schrodinger equation in one-dimension, free particle, particle in an infinite square well. [9]

Module-5: Modern Physics: Laser-Spontaneous and stimulated emission, Einstein's A and B coefficients, Population inversion, Light amplification, Basic laser action, Ruby and He-Ne lasers, Properties and applications of laser radiation, Nuclear Physics- Binding Energy Curve, Nuclear Force, Liquid drop model, Introduction to Shell model, Applications of Nuclear Physics, Concept of Plasma Physics, and its applications. [9]

Text books:

- 1: A. Ghatak, Optics, 4th Edition, Tata Mcgraw Hill, 2009
- 2: Mathew N.O. Sadiku, Elements of Electromagnetics, Oxford University Press (2001)
- 3: Arthur Beiser, Concept of Modern Physics, 6th edition 2009, Tata McGraw-Hill
4. F. F. Chen, Introduction to Plasma Physics and controlled Fusion, Springer, Edition 2016.

Reference books:

- 1: Fundamentals of Physics, Halliday, Walker and Resnick

Assessment Tool			% Contribution during CO Assessment				
Mid Sem Examination Marks					25		
End Sem Examination Marks					50		
Quiz					10+10		
Teacher's Assessment					5		
Assessment Compoents	CO1		CO2	CO3		CO4	CO5
Mid Sem Examination Marks	√		√	√			
End Sem Examination Marks	√		√	√		√	√
Quiz I				√		√	
Quiz II						√	√

Course Objectives

This course enables the students:

A.	To explain principles of physical optics and to have <u>basic</u> idea of <u>fiber optics</u> .
B.	To construct Maxwell's equations from basic principles and use <u>it</u> to solve electromagnetic plane wave equations.
C.	To distinguish between Newtonian Mechanics and <u>special</u> theory of relativity and develop the relationship of length contraction, time dilation <u>and</u> Einstein energy mass relation and to apply the concepts of <u>special</u> theory of relativity in various <u>field</u> of physics and engineering.
D.	To illustrate the phenomena of old quantum theory and derive <u>Heisenberg</u> uncertainty principle and <u>Schrodingers equations</u>
E.	To interpret basic lasing action, study various types of lasers, and <u>to</u> have <u>basic</u> idea of nuclear physics and plasma <u>physics</u>

Course Outcomes

After the completion of this course, students will be:

1.	Able to <u>analyse</u> the intensity variation of light due to Polarization, <u>interference and</u> diffraction.
2.	Able to formulate and solve the engineering problems on <u>electromagnetism</u>
3.	Able to explain <u>special</u> theory of relativity and apply its concepts in various fields of physics and engineering.
4.	Able to explain fundamentals of quantum mechanics and apply it to problems on bound <u>states</u>
5.	Able to explain <u>working</u> principle of lasers and <u>to</u> summarize <u>its</u> applications, understand <u>basic</u> idea of nuclear and plasma <u>physics</u>

Module-1:

Physical Optics: Polarization, Malus' Law, Brewster's Law, Double Refraction,

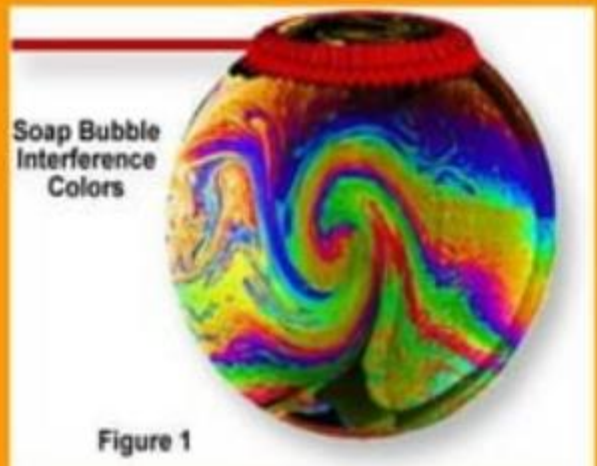
Interference in thin films (Parallel films),
Interference in wedge-shaped layers,
Newton's rings,

Fraunhofer diffraction by single slit, Double slit. Elementary ideas of fibre optics and application of fibre optic cables.

Physics

Wave Optics-Interference of Light

“Introduction and Thin Film Interference”



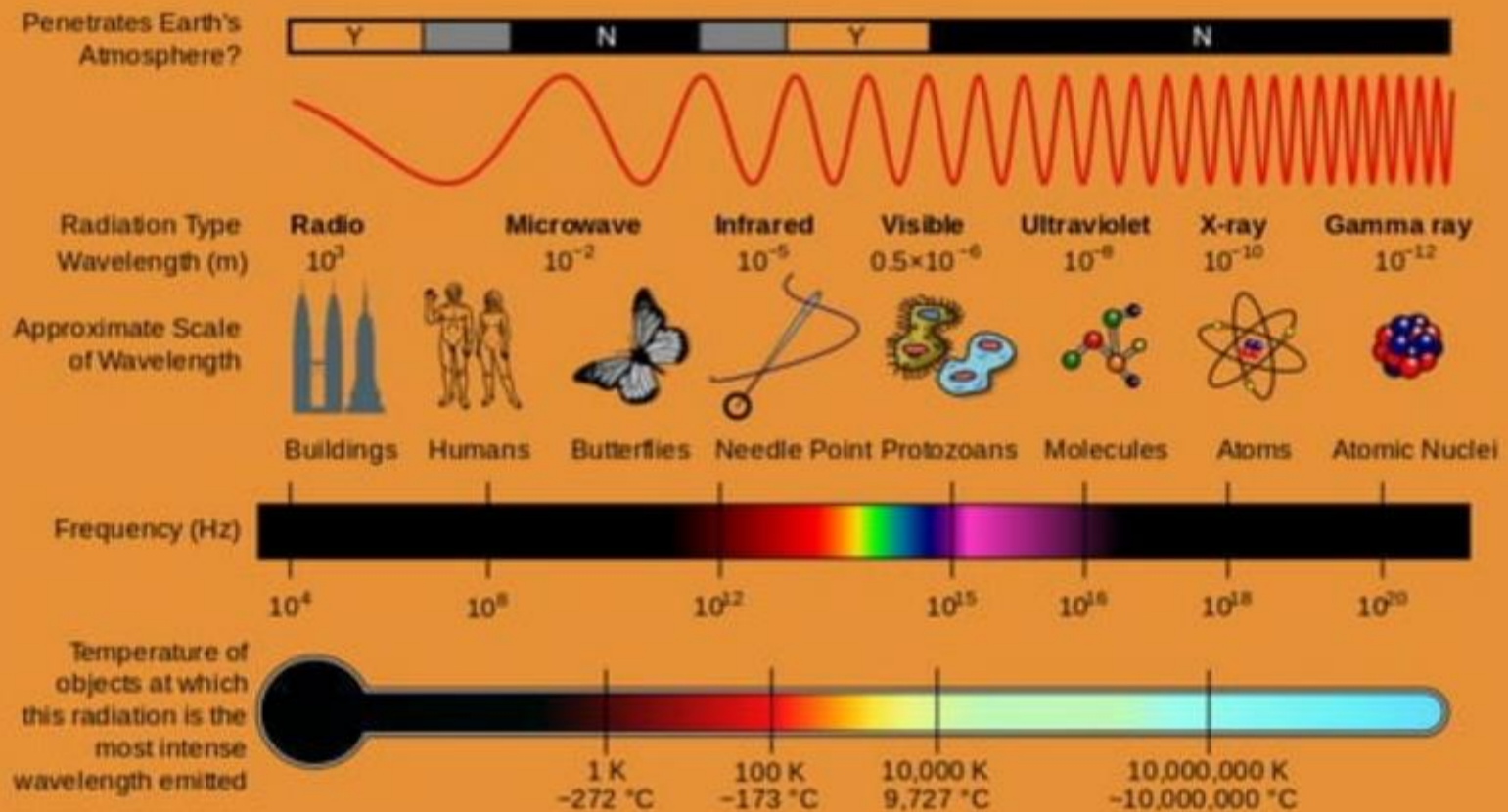
Contents

- Introduction
- Interference of Light and Principle of Superposition
- Young's Double Slit Experiment
- Conditions for Sustained Interference
- Coherent Sources
- Methods of Producing Coherent Sources
- Stoke's Law
- Thin Film Interference For Reflected and Transmitted Light

What is Light?

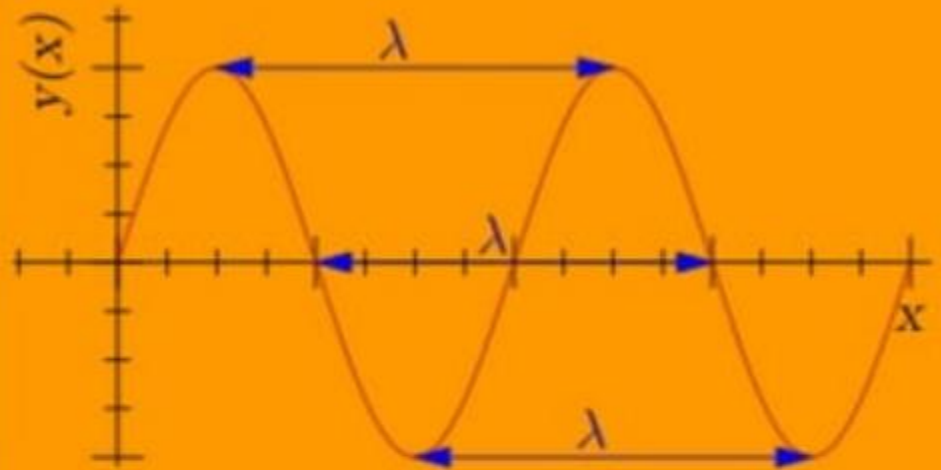


Light is an electromagnetic radiation refers to visible regions of electromagnetic spectrum corresponding to the wavelength range of 380nm to 750nm which has transverse vibrations.

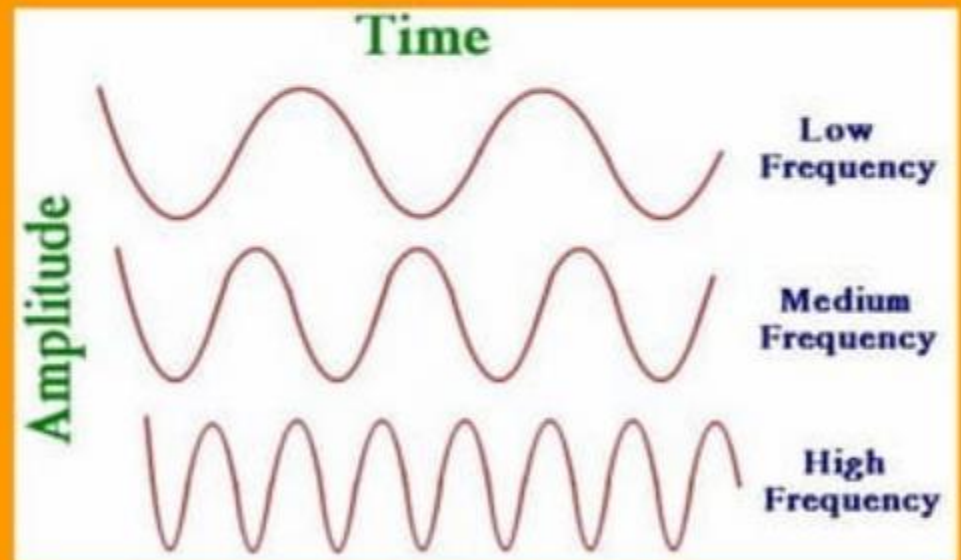


Basic Definitions

The Wavelength of a sin wave, λ , can be measured between any two points with the same phase, such as between crests, or troughs, as shown.



The frequency, f , of a wave is the number of waves passing a point in a certain time. We normally use a time of one second, so this gives frequency the unit hertz (Hz), since one hertz is equal to one wave per second.



Path Difference and Phase Difference

At time $t = 0$

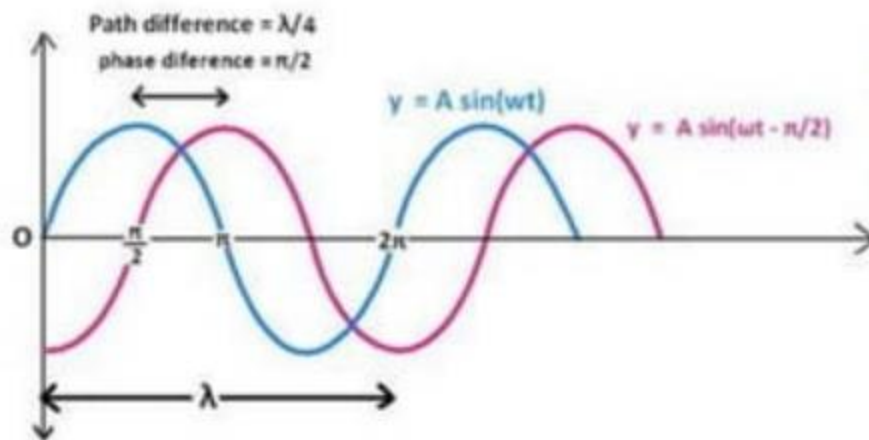
Blue wave displacement = 0

Red wave displacement = $-A$

At $\pi/2$

Blue wave has maximum displacement = $+A$

Red wave displacement = 0



Blue wave is leading by a phase difference of $\pi/2$ and path difference of $\lambda/4$

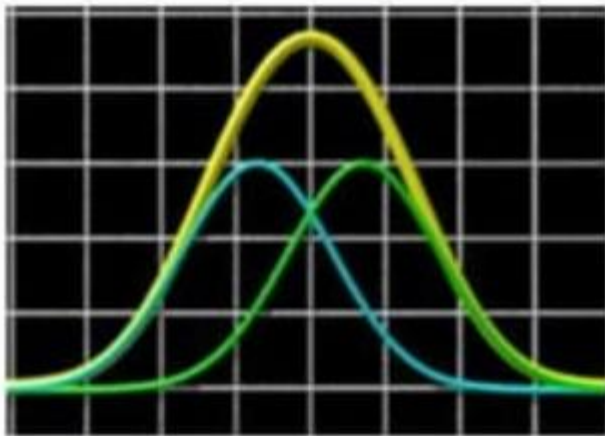
One oscillation is completed in 2π radians which is equivalent to wavelength λ

(Path difference of one wavelength (λ) is equal to phase difference of 2π radians)

Principle of Superposition

“Whenever two or more waves superimpose in a medium, the total displacement at any point is equal to the vector sum of individual displacement of waves at that point”

$$y = y_1 + y_2 = a \sin(\omega t) + a \sin(\omega t + \phi)$$



If $Y_1, Y_2, Y_3 \dots$ are different displacement vectors due to the waves 1, 2, 3 ... acting separately then according to the principle of superposition the resultant displacement is given by

$$Y = Y_1 + Y_2 + Y_3 + \dots$$

Introduction...

Optics

It is a branch of Physics in which we study the behaviour and properties of light. This study also includes the interaction of light with matter and the construction of instruments that are used to detect it.

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graph TD; A[Optics] --> B[Geometric Optics or Ray Optics]; A --> C[Physical Optics or Wave Optics]; A --> D[Quantum Optics]
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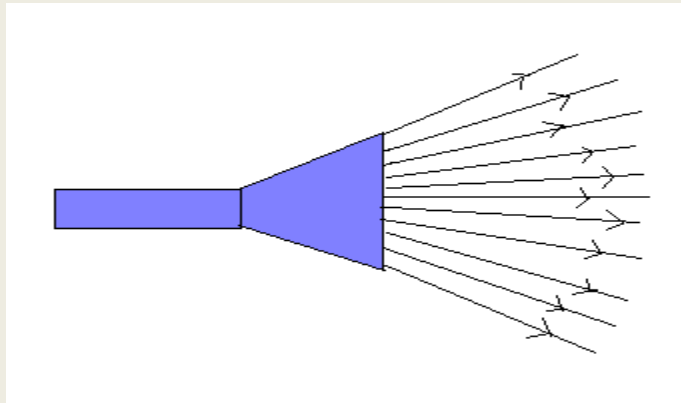
Geometric Optics
or Ray Optics

Physical Optics
or Wave Optics

Quantum Optics

Geometric Optics or Ray Optics

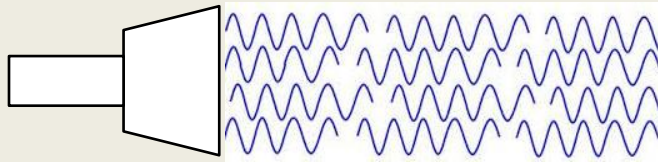
- Propagation of light is represented by 'Rays'.



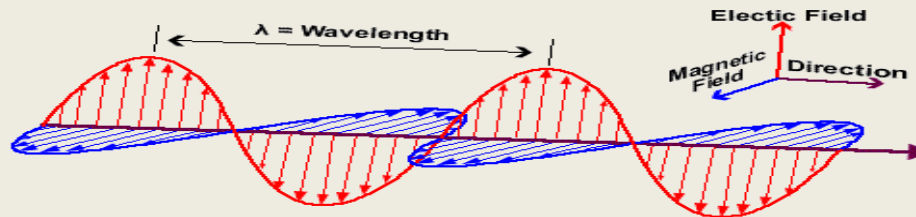
- In a homogeneous medium the ray paths are straight line.
- Ray bends or changes path, at the interface between two dissimilar media.
- It can easily explain the phenomena of Reflection and Refraction of Light but it fails to explain Interference, Diffraction, Polarization etc.

Physical Optics or Wave Optics

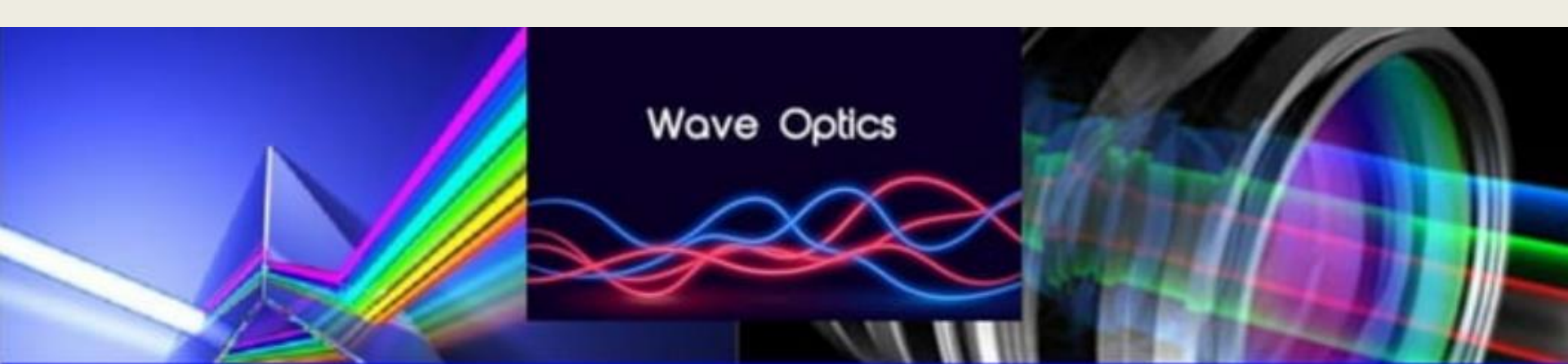
- A beam of light comprises of large number of propagating waves.



- These are electromagnetic waves.



- The phenomena like Interference, Diffraction, Polarization etc can be explained by Wave Optics.



Wave Optics

wave optics, is the branch of **optics** that studies interference, diffraction, polarization, and other phenomena for which the ray approximation of geometric **optics** is not valid.



Quantum Optics

- Propagation of light is described in terms of large number of tiny particle-like packets of optical energy called “photons”, which travel with the speed of light ‘c’
- The energy of a photon corresponding to light of a particular colour, with wavelength λ , is given by

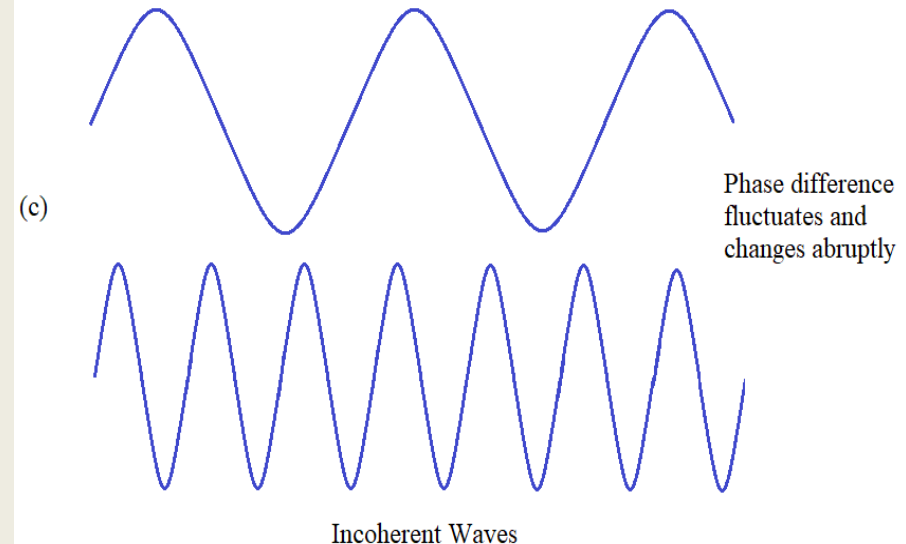
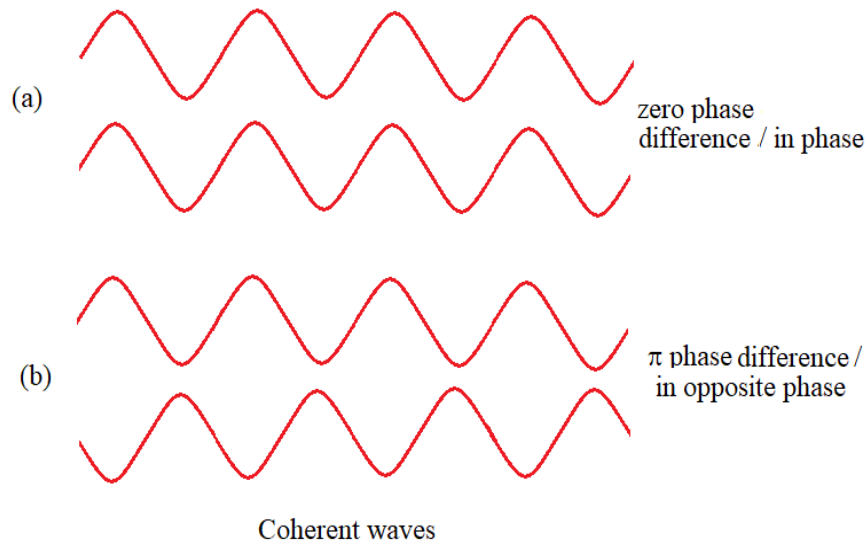
$$E = \frac{hc}{\lambda} = h\nu$$

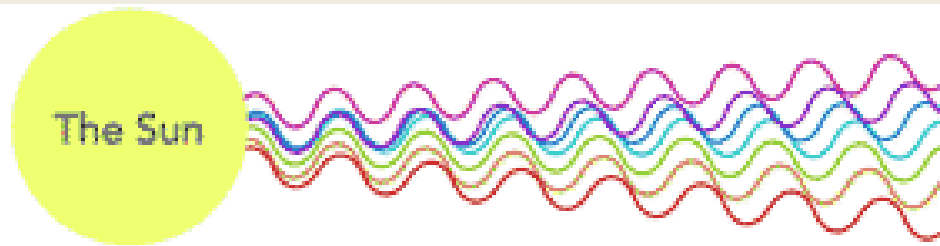
where h = planck's constant = 6.626×10^{-34} J-s

Coherent Sources

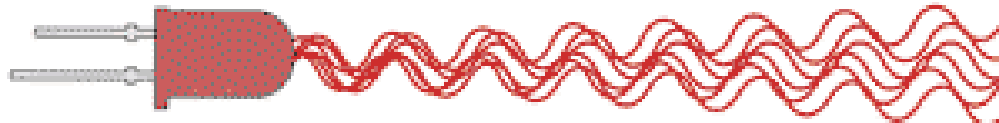
Two sources are said to be coherent if they emit light waves of the

- Same frequency
- Nearly the same amplitude
- Phase difference between them remains constant with respect to time.

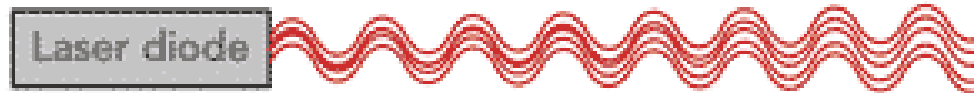




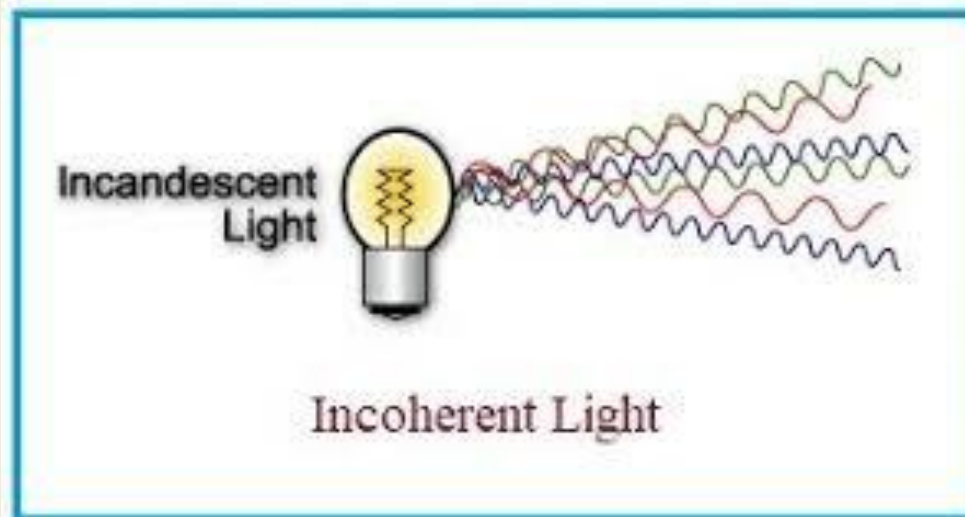
White light - different frequencies and phase



Incoherent light - same frequency and different phase

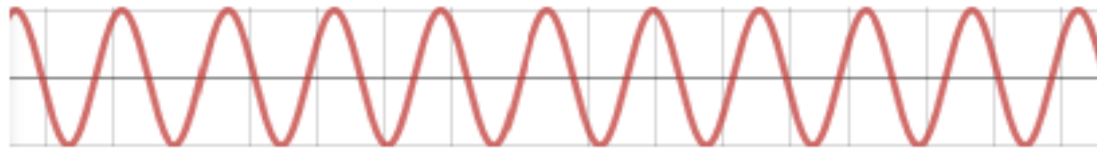


Coherent light - same frequency and phase

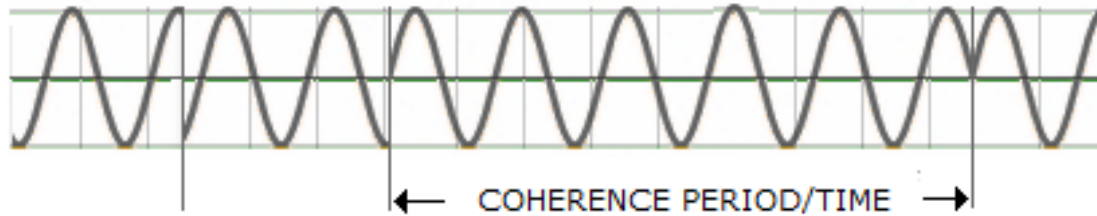


Incoherent Light

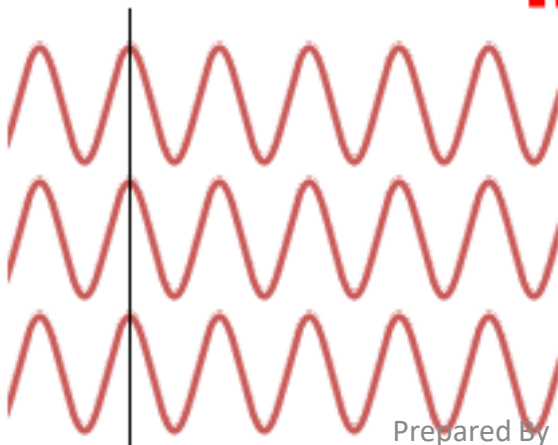
temporally coherent wave (monochromatic)



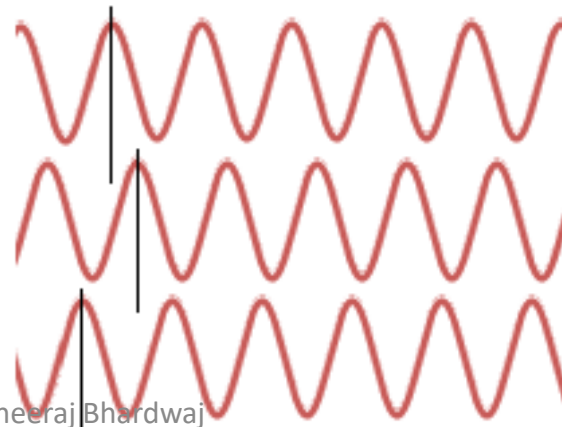
temporally incoherent

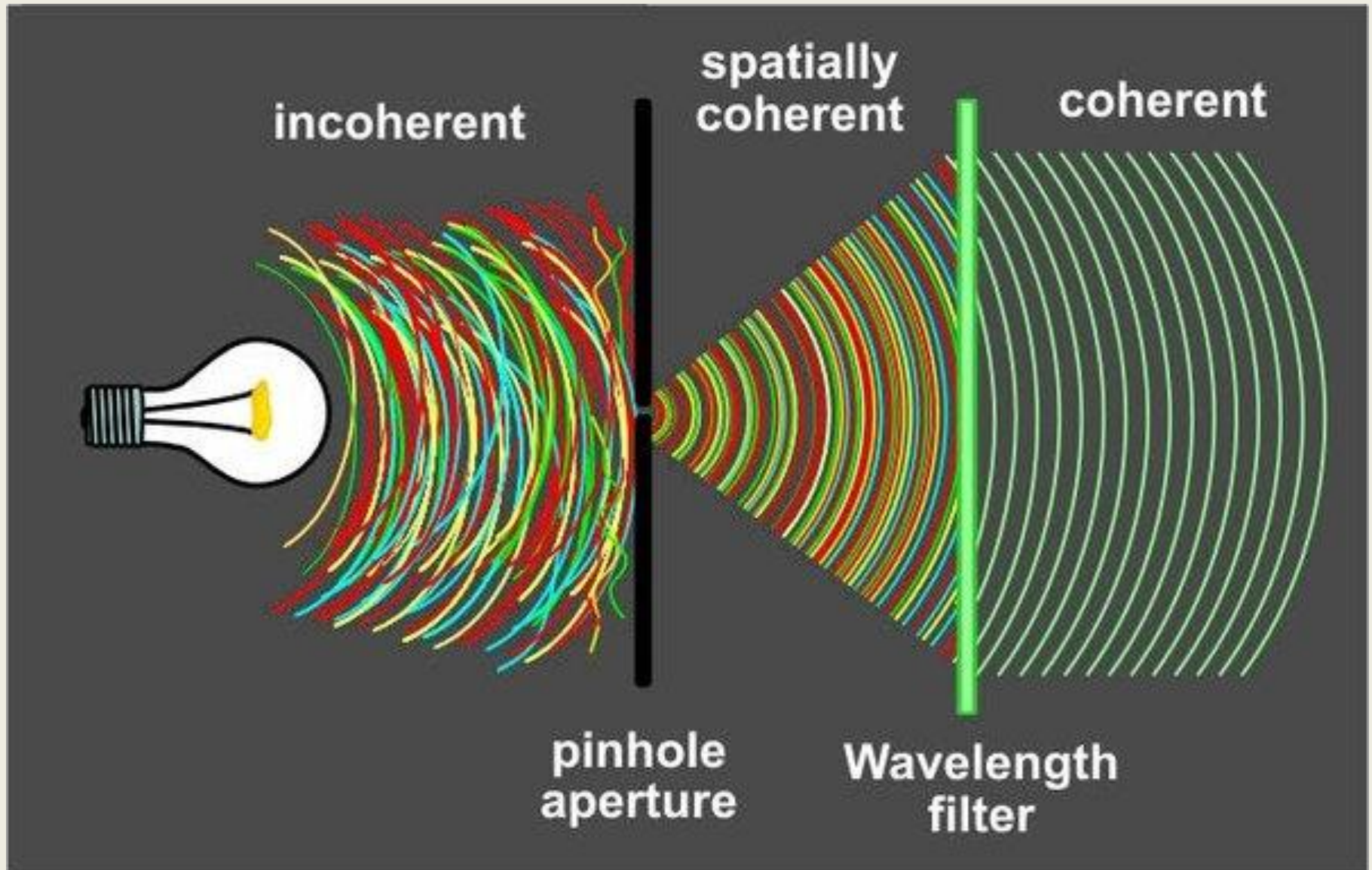


SPATIALLY COHERENT WAVES



SPATIALLY INCOHERENT



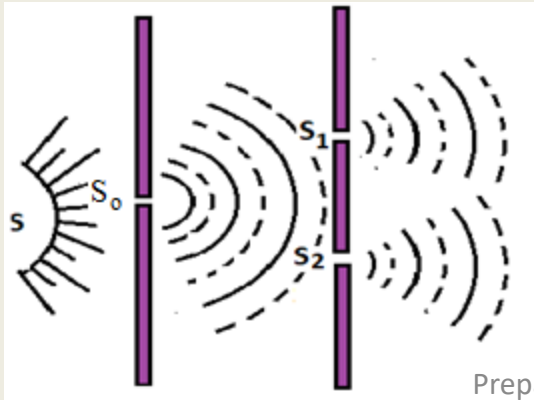


Two Methods of Obtaining Coherent Sources

Division of wave front

The wave front from a single monochromatic source is divided either, into two to form two virtual images of the source, or into one virtual and one real source using reflection or refraction.

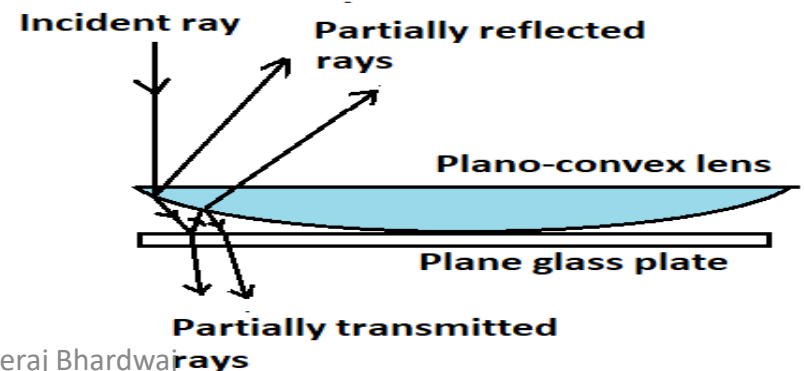
Examples: Fresnel's biprism, Lyod's mirror, Young's double slit etc.



Division of amplitude

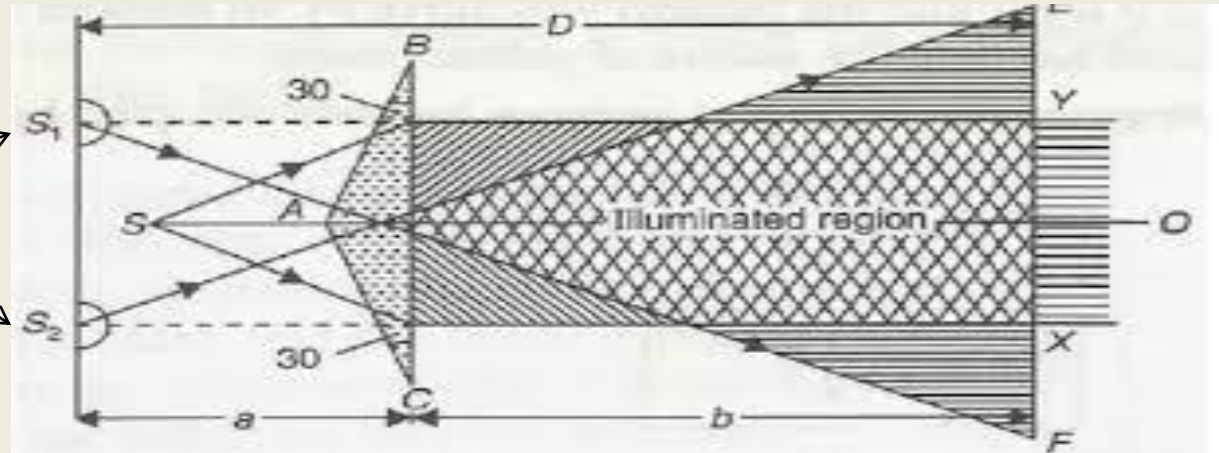
The amplitude of wave emitted by an extended source of light is divided in two parts by partial refraction and partial reflection.

Examples: Thin film Interference, Newton's rings, Michelson's interferometer etc.

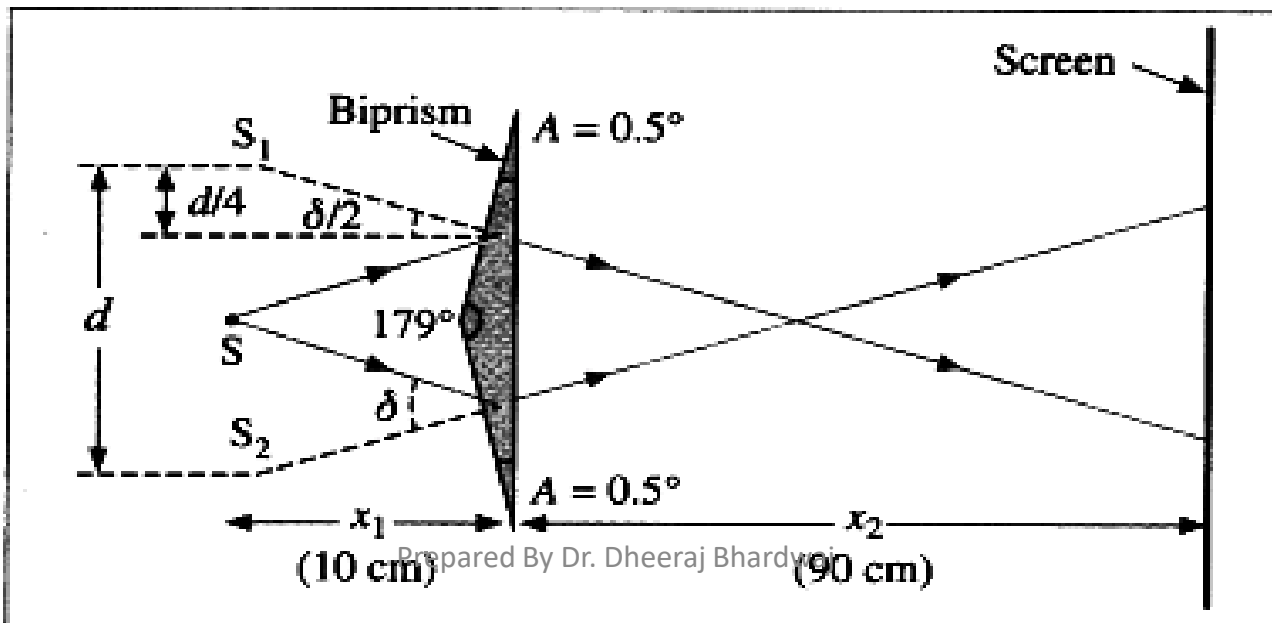


Coherent Sources by division of Wavefront

Two virtual images of the Source



Fresnel Biprism Experiment



Interference as Superposition of Waves in Space

Interference of Light:

When two waves of same frequency and nearly same amplitude travel in approximately in same direction and have phase difference that remains constant with respect to time, the intensity of light is redistributed in space. This non uniform distribution of light intensity due to superposition of two waves is called **interference of light**.

Principle of Superposition:

Whenever two or more waves superimpose (meet), the instantaneous total displacement at any point is equal to the vector sum of the individual displacements of wave at that point.

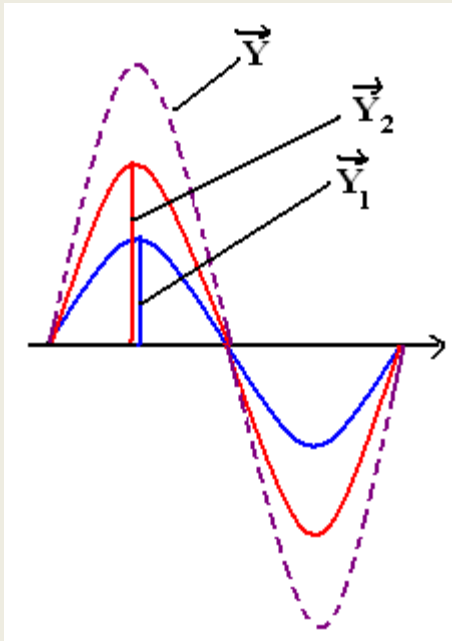
If $\vec{E}_1, \vec{E}_2, \vec{E}_3 \dots$ are the disturbances produced by the individual waves, then the resultant disturbance will be

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 \dots$$

Principle of Superposition....

Case I: When the displacement of two waves are in the same direction (i.e. same phase) then resultant displacement is

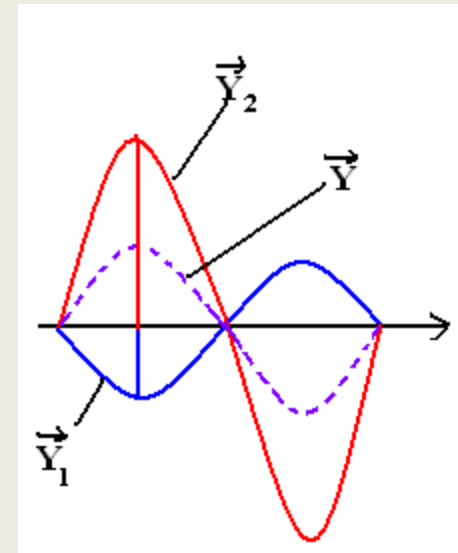
$$\vec{Y} = \vec{Y}_1 + \vec{Y}_2$$



Case II: When the displacement due to two waves are mutually in opposite direction (i.e. opposite phase) then resultant displacement will be

$$\vec{Y} = \vec{Y}_2 - \vec{Y}_1$$

Here $Y_2 > Y_1$

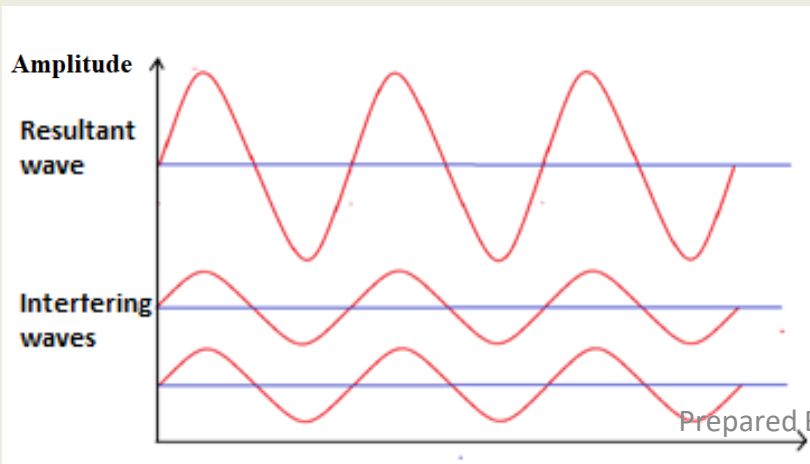


Interference of Light

Interference

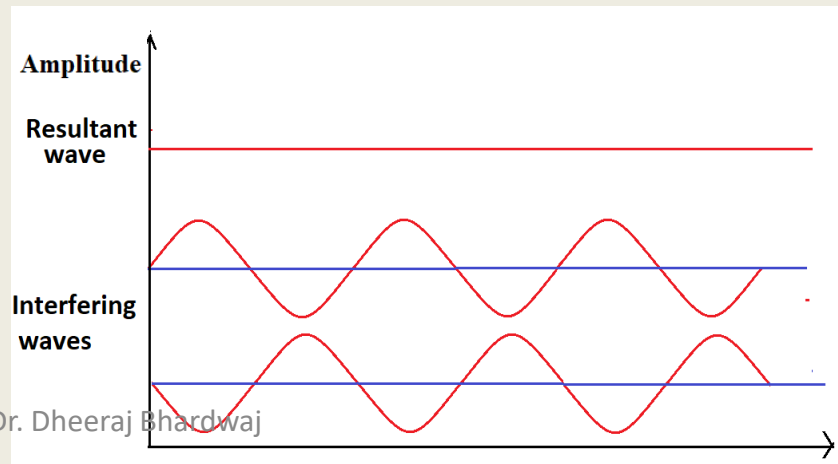
Constructive Interference

- The amplitude of resultant wave is maximum.
- The two waves meet in same phase.
- The phase difference betⁿ two waves is $\delta = 2n\pi$ when $n=0,1,2,3,\dots$
- The path difference betⁿ two waves is $\Delta = n\lambda$ where $n=0,1,2,\dots$

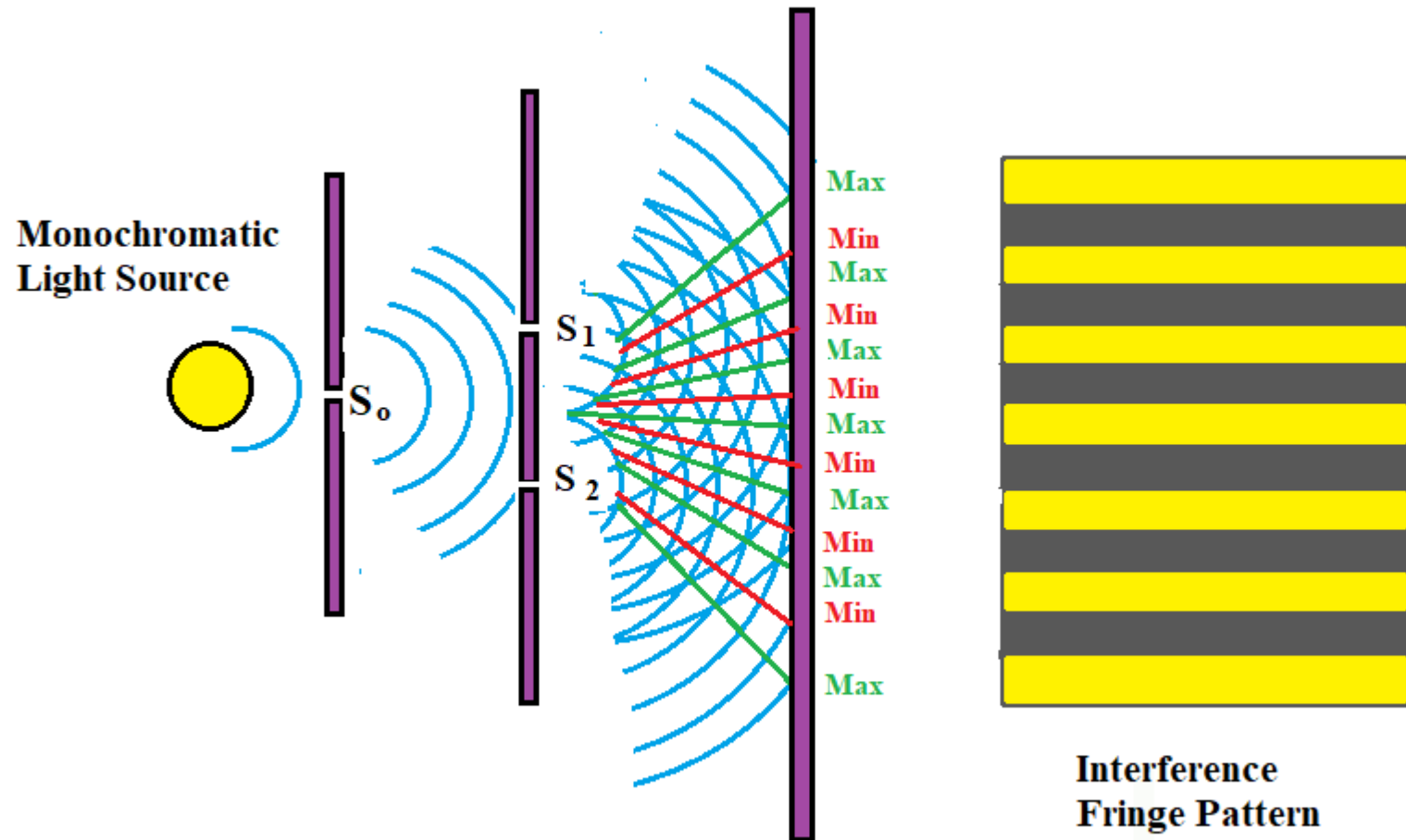


Destructive Interference

- The amplitude of resultant wave is minimum.
- The two waves meet in opposite phase.
- The phase difference betⁿ two waves is $\delta = (2n-1)\pi$ when $n=1,2,3,\dots$
- The path difference betⁿ two waves is $\Delta = (2n-1)\lambda/2$ where $n=1,2,\dots$



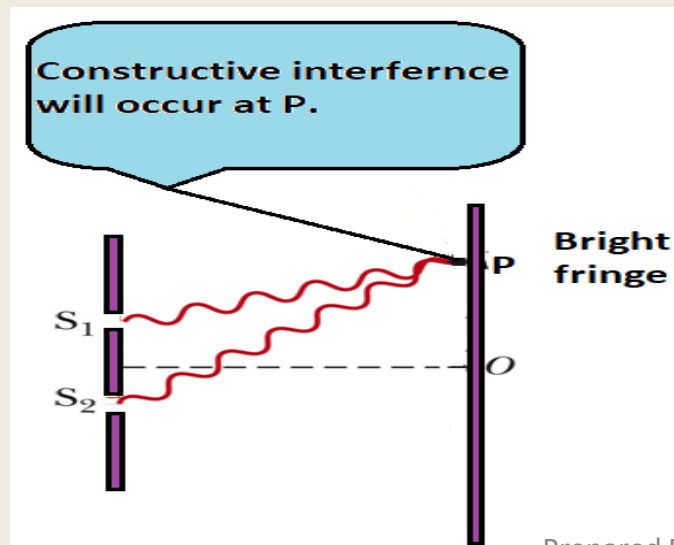
Young's Double Slit Experiment



Young's Double Slit Experiment contd..

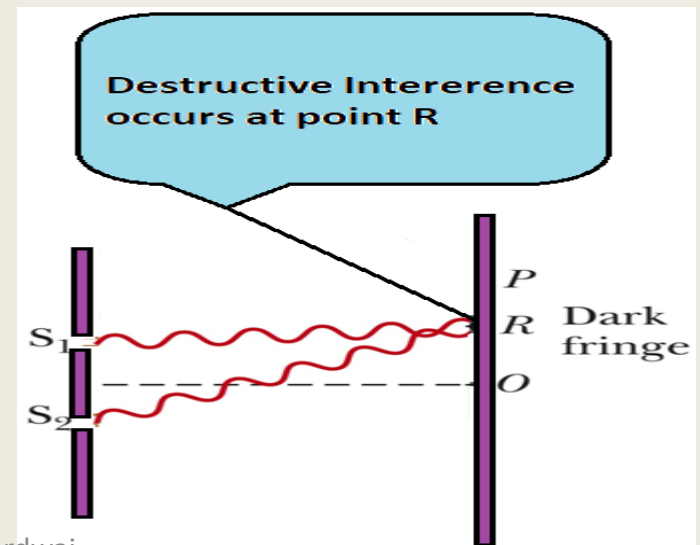
Constructive Interference

If the two waves reaching at point P are in phase i.e. the phase difference between them is equal to $2n\pi$ or path difference between them is equal to $n\lambda$, then a bright fringe will form there.



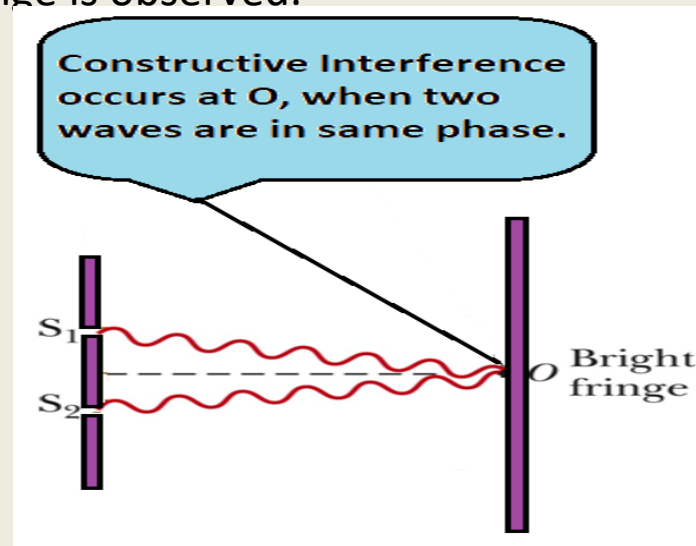
Destructive Interference

If the two waves reaching at point P are out of phase i.e. the phase difference between them is equal to $(2n-1)\pi$ or path difference between them is equal to $(2n-1)\lambda/2$, then a dark fringe will form there.



Young's Double Slit Experiment contd..

Constructive interference occurs at point O. The two waves travel the same distance. Therefore, they arrive in phase. As a result, constructive interference occurs at this point and a bright fringe is observed.

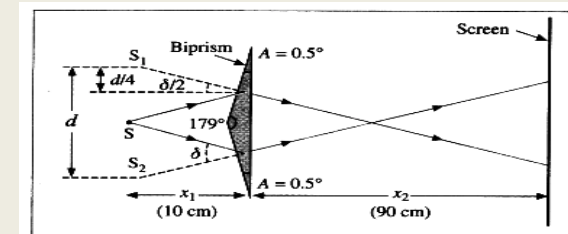


Let a_1 and a_2 are amplitudes of these two waves of same angular frequency ω and δ is the phase difference between the two waves reaching a point on screen at any instant. The intensity of resultant wave on the screen is given by

$$I = a_1^2 + a_2^2 + 2a_1a_2 \cos \delta$$

Conditions for Sustained Interference

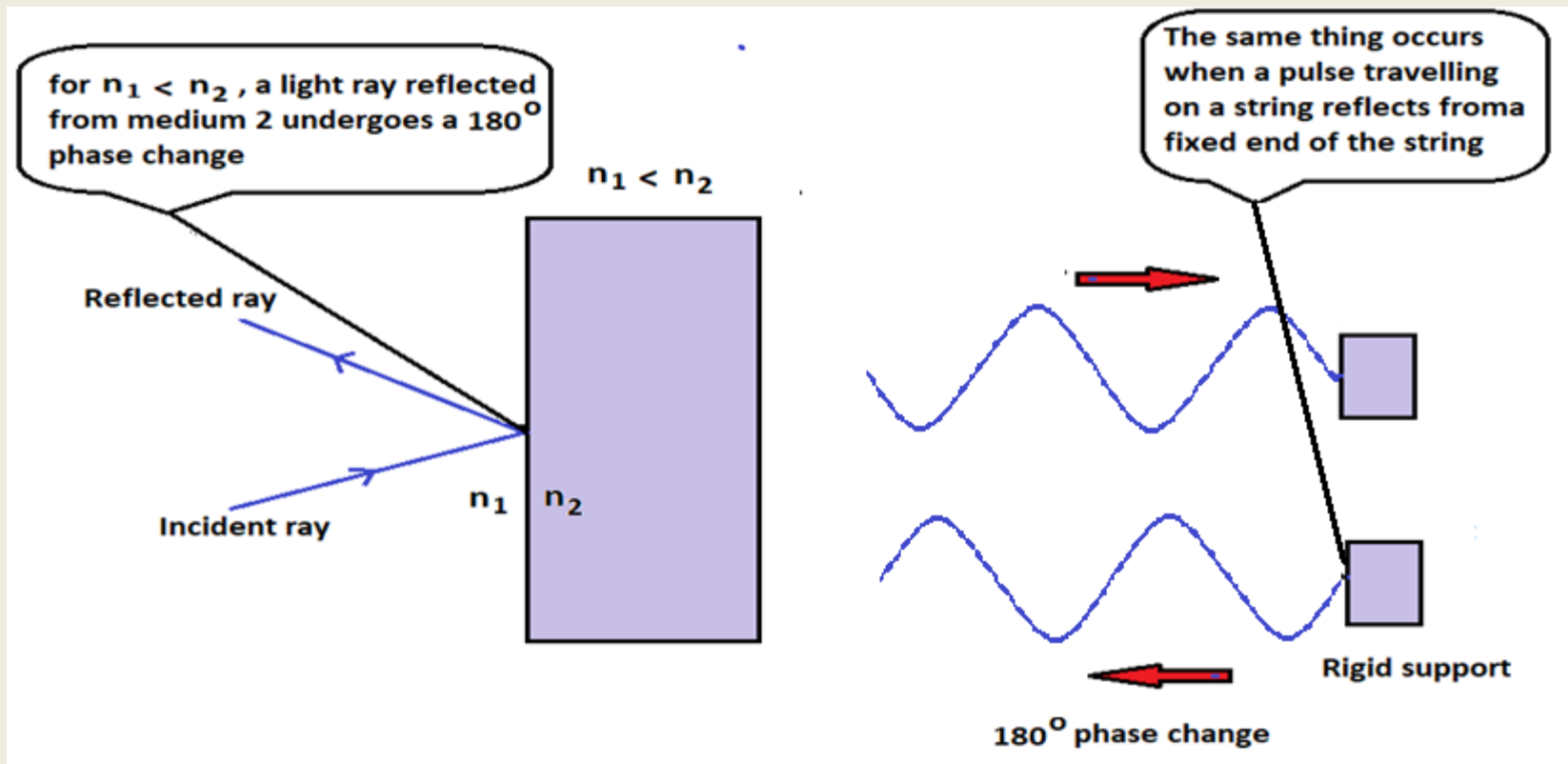
- The waves from the two sources must be of the same frequency or wavelength.
- The two light waves must be coherent.
- The two light waves should propagate in the same direction.
- The amplitude of both wave should be nearly equal so that good contrast and distinct interference fringe pattern will be obtained.
- Separation between the two coherent light sources S_1 and S_2 should be as small as possible.



- The distance of the screen from the two coherent sources must be large.
- If waves are polarized, then polarization state of both waves should be same.

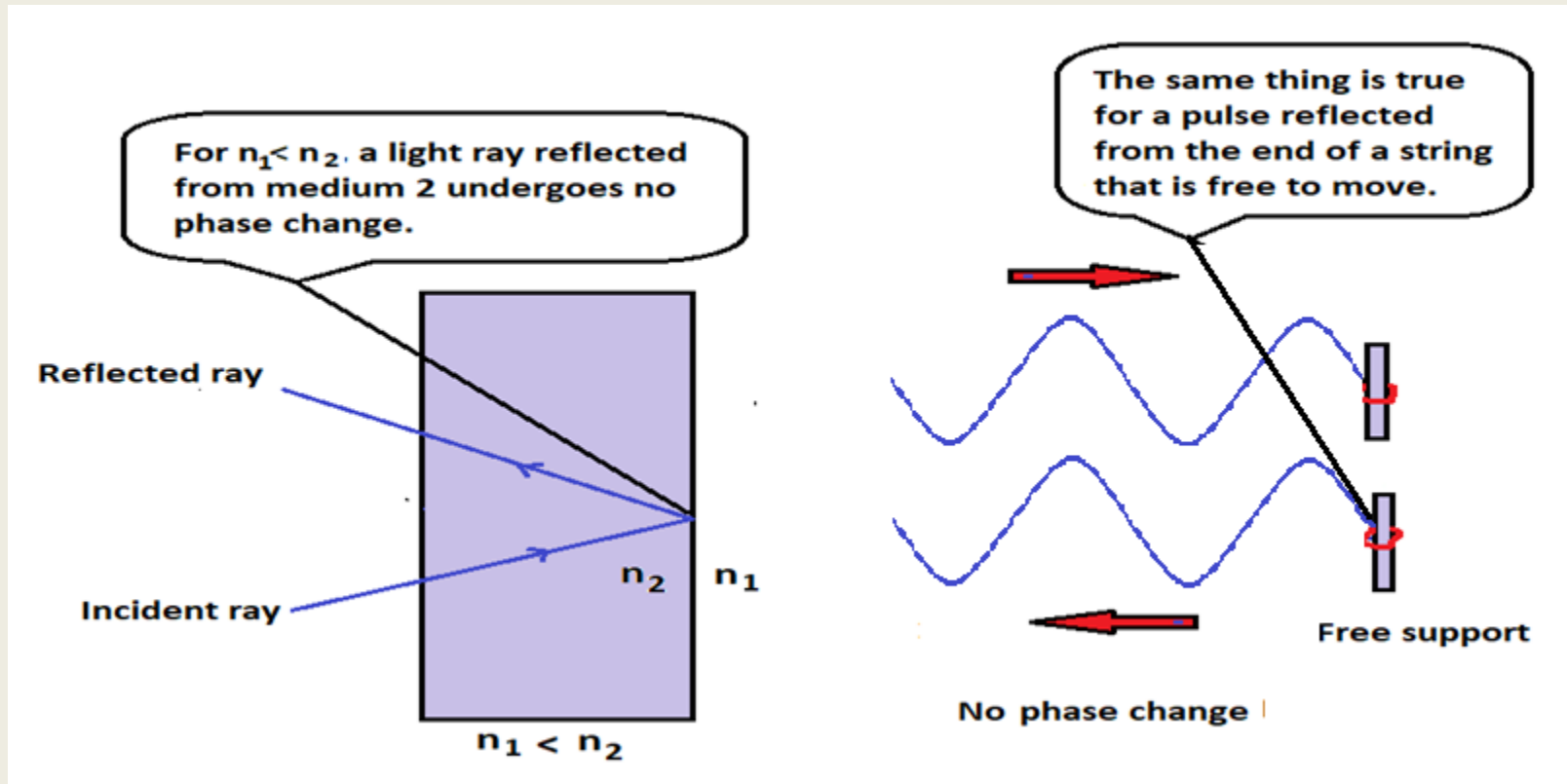
Phase Changes Due To Reflection (Stoke's Law)

An electromagnetic wave undergoes a phase change of 180° upon reflection from a medium of higher index of refraction than the one in which it was traveling.



Phase Changes Due To Reflection (Stoke's Law) contd...

- There is no phase change when the wave is reflected from a boundary leading to a medium of lower index of refraction.



Interference in Thin Film

The various colors observed when white light is incident on such films result from the interference of waves reflected from the two surfaces of the film.



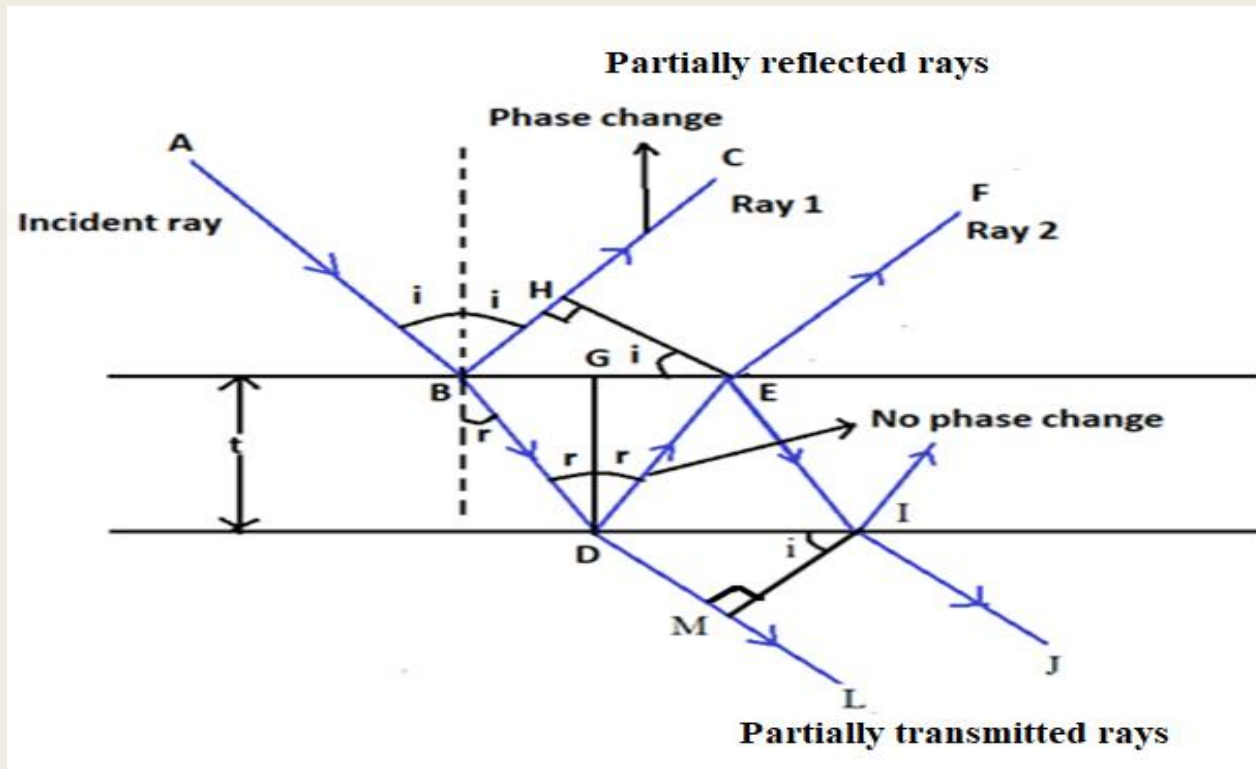
Thin film interference in a soap bubble. Color varies with film thickness.



A colourful interference pattern is observed when light is reflected from the top and bottom boundaries of a thin oil film. The different bands form as the film's thickness diminishes from a central runoff-point.

Interference in Thin Film

The formation of interference pattern is due to the superposition of coherent rays produced by the division of amplitude.



Thickness of film = t

Refractive index of film = μ

Prepared By Dr. Dheeraj Bhardwaj

Interference in Thin Films by Reflection.....

The path difference between the light ray 1 and ray 2

Path difference =>

$$= [\{\mu(BD + DE) \text{ in film}\} - (BH \text{ in air})] \dots(1)$$

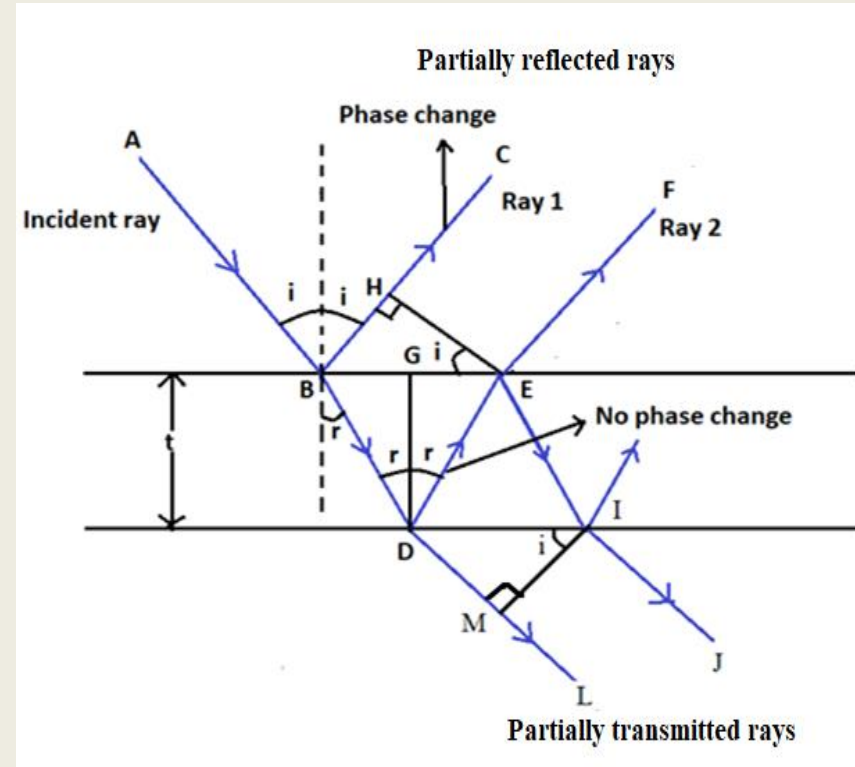
From $\triangle BDG$

$$\cos r = \frac{DG}{BD} = \frac{t}{BD} \Rightarrow BD = \frac{t}{\cos r} \dots(2)$$

Similarly, from $\triangle DEG$

$$\cos r = \frac{DG}{DE} = \frac{t}{DE} \Rightarrow DE = \frac{t}{\cos r} \dots(4)$$

$$BD = DE = \frac{t}{\cos r}$$



Thickness of film = t

Refractive index of film = μ

Interference in Thin Films by Reflection.....

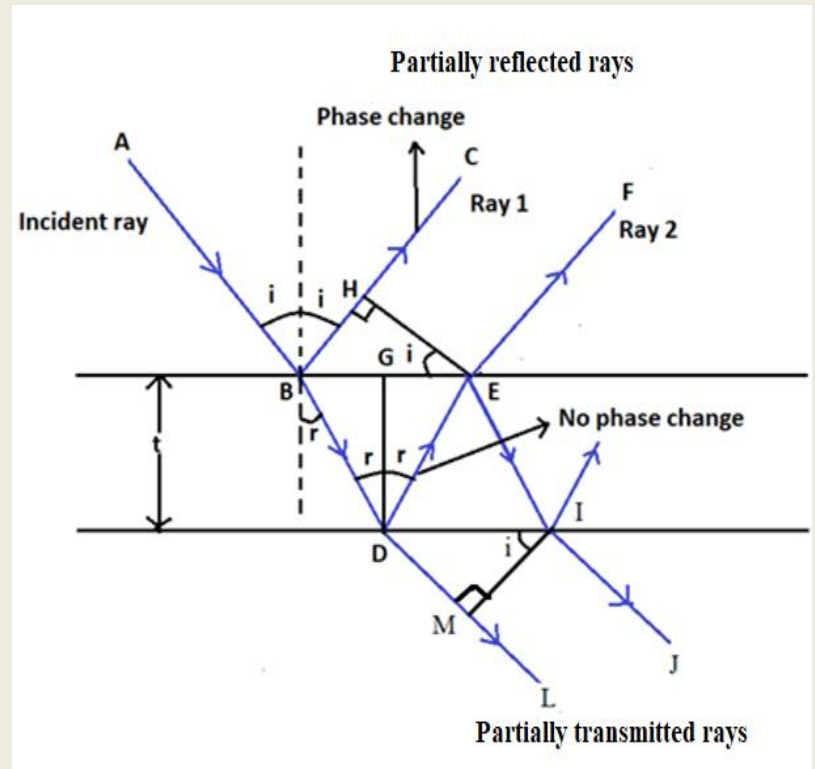
$$\begin{aligned}
 &= \frac{2\mu t}{\cos r} - 2\mu t \cdot \frac{\sin^2 r}{\cos r} \\
 &= \frac{2\mu t}{\cos r} (1 - \sin^2 r) \\
 &= \frac{2\mu t}{\cos r} \cos^2 r
 \end{aligned}$$

$$\Delta = 2\mu t \cos r \quad \dots(11)$$

At point B the light ray (1) is reflected at the surface of thin film (denser medium). So the light ray (1) undergoes a phase change π or an additional path difference $\lambda/2$.

Effective path difference

$$\Delta_{eff} = 2\mu t \cos r - \frac{\lambda}{2}$$



Thickness of film = t

Refractive index of film = μ

Interference in Thin Films by Reflection.....

Constructive interference (or Bright fringe):

General condition; path difference = $n\lambda$

$$2\mu t \cos r - \frac{\lambda}{2} = n\lambda$$

$$\Rightarrow 2\mu t \cos r = n\lambda + \frac{\lambda}{2}$$

$$\Rightarrow 2\mu t \cos r = \frac{(2n+1)\lambda}{2} \quad \dots(13)$$

Where $n = 0, 1, 2, 3, 4, \dots$

Destructive interference (or Dark fringe):

General condition; path difference

$$= \frac{(2n-1)\lambda}{2}$$

$$2\mu t \cos r - \frac{\lambda}{2} = \frac{(2n-1)\lambda}{2}$$

$$\Rightarrow 2\mu t \cos r = n\lambda \quad \dots(14)$$

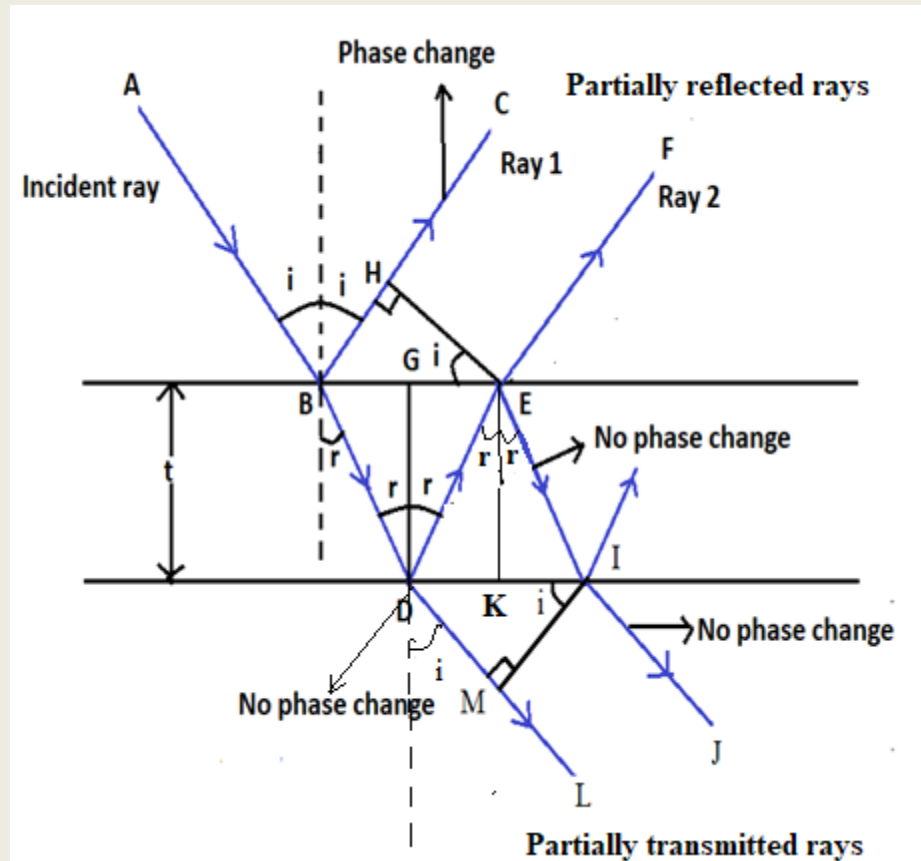
Where $n = 0, 1, 2, 3, 4, \dots$

Interference in Thin Films by Transmission

Here ray DL and DEIJ are transmitted rays. These two light rays are parallel and superimpose and produce interference. The intensity of interference fringe depends up on the path difference between the two rays.

As none of the transmitted rays is reflected by optically denser medium so, there will be no phase change and the path difference between the two transmitted rays will be

$$\Delta_{eff} = 2\mu t \cos r \quad \dots(15)$$



Thickness of film = t

Refractive index of film = μ

Interference in Thin Films by Transmission...

Constructive interference (or Bright fringe):

General condition; path difference = $n\lambda$

$$2\mu t \cos r = n\lambda \quad \dots(16)$$

Where $n = 0, 1, 2, 3, 4, \dots$

Destructive interference (or Dark fringe):

General condition; path difference

$$= \frac{(2n-1)\lambda}{2}$$

$$2\mu t \cos r = \frac{(2n-1)\lambda}{2} \quad \dots(17)$$

Where $n = 1, 2, 3, 4, \dots$

NOTE:- “ INTERFERENCE IN REFLECTED AND TRANSMITTED LIGHT ARE COMPLEMENTARY TO EACH OTHER”

Interference in Thin Films by Reflection

Constructive interference (or Bright fringe):

$$2\mu t \cos r = \frac{(2n+1)\lambda}{2}$$

Where $n = 0, 1, 2, 3, 4, \dots$

Destructive interference (or Dark fringe):

$$2\mu t \cos r = n\lambda$$

Where $n = 0, 1, 2, 3, 4, \dots$

Interference in Thin Films by Transmission

Constructive interference (or Bright fringe):

$$2\mu t \cos r = n\lambda$$

Where $n = 0, 1, 2, 3, 4, \dots$

Destructive interference (or Dark fringe):

$$2\mu t \cos r = \frac{(2n-1)\lambda}{2}$$

Where $n = 1, 2, 3, 4, \dots$

➤ **INTERFERENCE IN REFLECTED AND IN TRANSMITTED LIGHT ARE COMPLEMENTARY TO EACH OTHER** because the condition for maxima equation (13) and minima equation (14) in reflected light are just reversed to the conditions in transmitted light [equations (16) and (17)].

➤ Thus the fringe of a certain order which appears bright in reflected light will appear dark in transmitted light and vice versa.

➤ In case of transmitted light, the interference fringes obtained are less distinct because the difference in amplitude between transmitted rays DL and DEIJ is very large (not comparable to zero).



Colors of thin films:

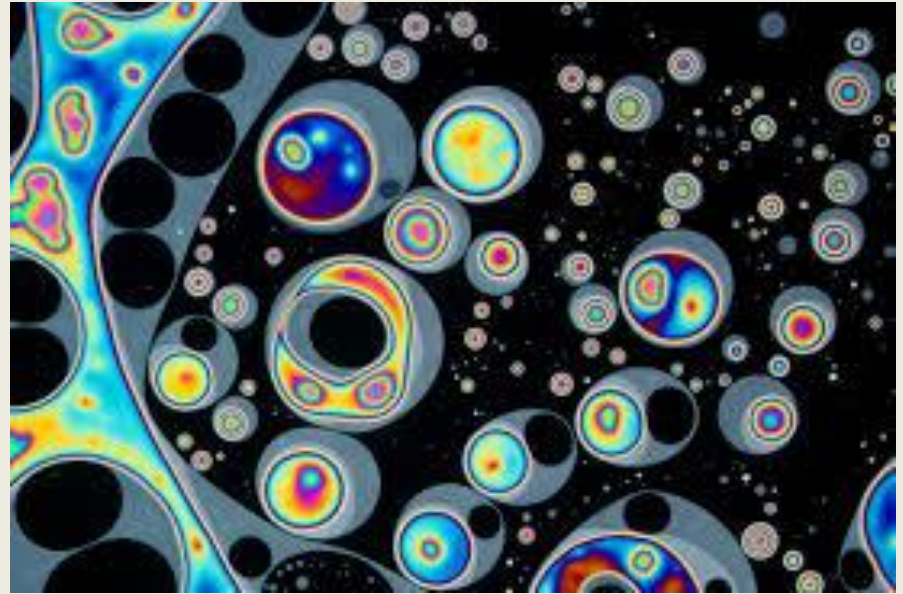


When a thin film is illuminated by monochromatic light and seen in reflected light, it will appear bright if $2\mu t \cos r = (2n+1) \lambda/2$ and dark if $2\mu t \cos r = n\lambda$. However, the film is illuminated by white light, the film shows different colors.

The eye looking at the film receives the waves of light reflected from the upper and lower surfaces of the film. For a thin film, these rays are very close to each other. The path difference between interfering rays is $(2\mu t \cos r - \lambda/2)$.

The path difference depends upon t , the thickness of the film, and upon r which depends upon the inclination of the incident rays. Now white light consists of a continuous range of wavelengths. For a particular value of t and r i.e. at a particular point of the film and for a particular position of the eye, the waves of only certain wavelengths satisfy the condition of constructive interference. Therefore, only those colors will be present in the reflected system with maximum intensity. The other neighbouring wavelengths will be present with less intensity. There will also be certain wavelengths which satisfy the condition of minima. Such wavelengths of colors will be absent from the reflected light. As a result, the point of the film will appear colored.

Interference due to an infinitely thin film :-

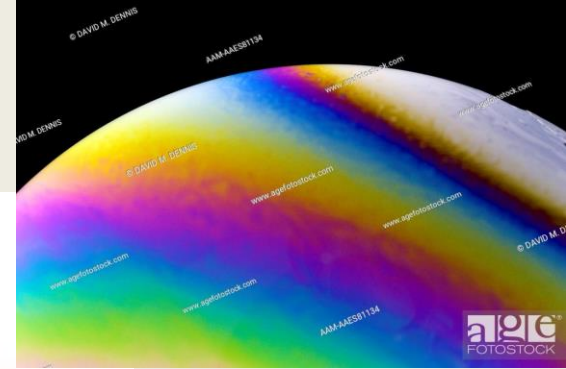
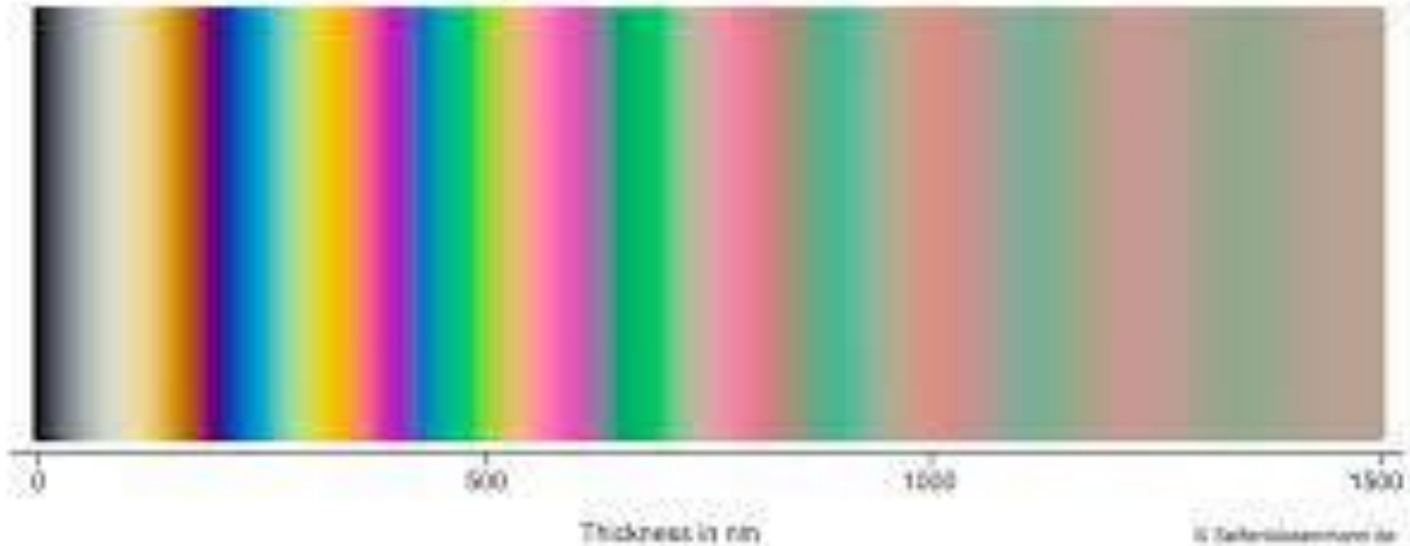


When the thickness (t) of the film is small compared to the wavelength of light ($\lambda \gg t$). The effective path difference between the interfering waves in reflected light for a film is $2ut \cos r - \lambda/2$. When the film is excessively thin such that its thickness t is very small ($t \approx 0$) compared to the wavelength of light then $2ut \cos r$ is almost zero. Hence the effective path difference becomes $\lambda/2$. This is condition for minimum intensity. **Hence every wavelength will be absent and the film will appear black in reflected light and it will appear bright in transmitted light.**

Interference in a thick film :-

The colors of a soap film

assuming sunlight with normal angle of incidence



When the thickness of the film is large compared to the wave length of light ($t \gg \lambda$), the path difference at any point of the film will be large. In this case the condition of constructive interference ($2\mu t \cos r = (n+1/2) \lambda$), at a given point is satisfied by large number of wave length with the value of n different for different colours. At the same point the condition of destructive interference ($2\mu t \cos r = n\lambda$) is also satisfied for another set of large number of wavelength. **Thus in the case of a thick film illuminated by white light, the colors are not observed in the reflected light and it appears uniformly illuminated.**

Necessity of an Extended Light Source

An extended source is necessary to enable the eye to see whole of the film and observed the interference pattern due to whole film. Let us consider a thin film and a narrow source of light at S (fig.a) then for each incident ray E_1, E_2 we get a pair of parallel interfering rays. The incident ray produces interference fringes because $E_{1r}, E_{1r'}$ reach the eye where as the incident ray E_2 meet the surface at some different angle and is reflected along $E_{2r} E_{2r'}$. Here $E_{2r} E_{2r'}$ do not reach the eye. Therefore, the position A of the film is visible.

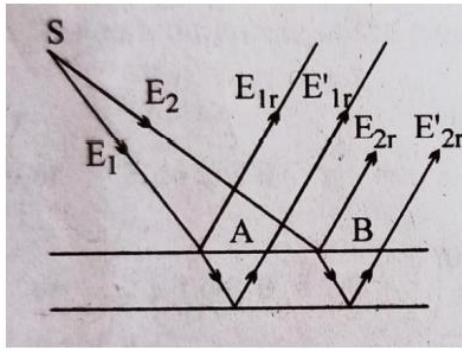


Fig - a

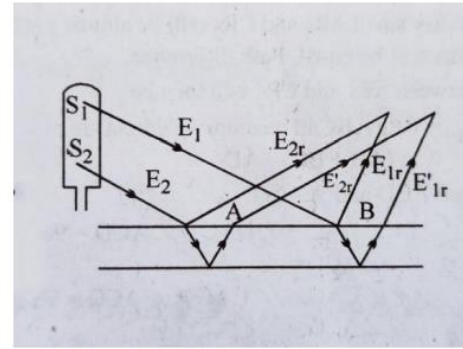


Fig - b

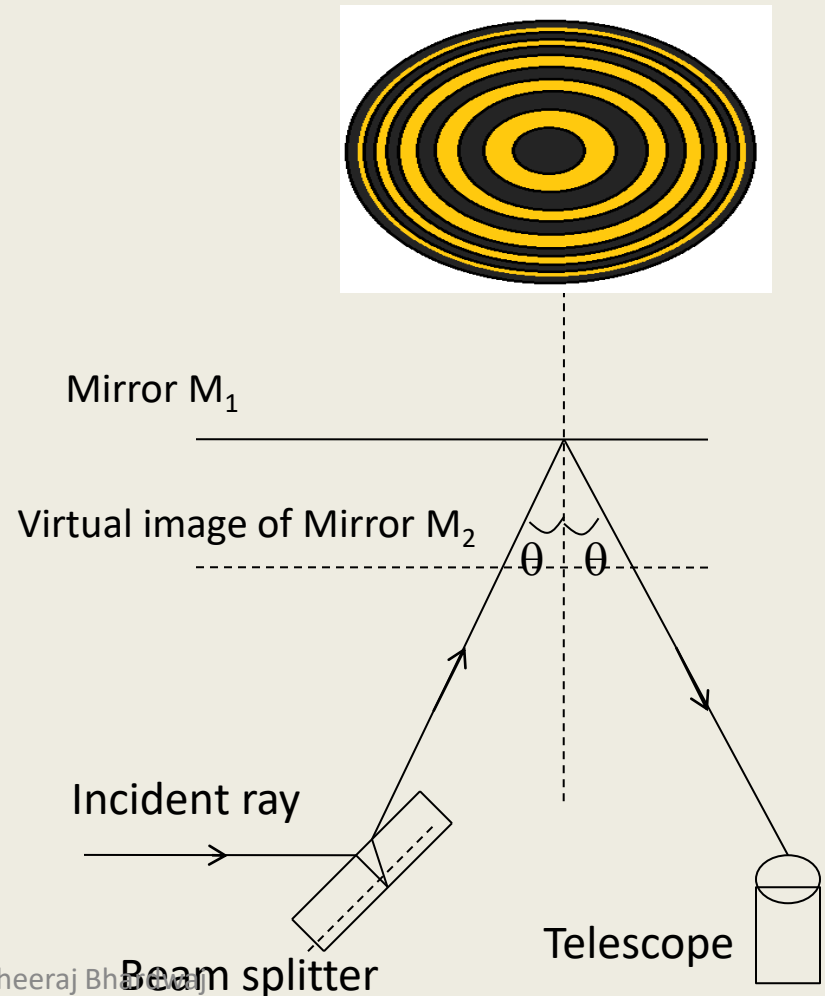
If extended source of light is used (fig.b) the incident ray E_1 after reflection from the upper and the lower surface of the film emerges as $E_{1r}, E_{1r'}$ which reach the eye. Also incident ray E_2 from some other point of the source after reflection from the upper and lower surface of the film emerges as $E_{2r}, E_{2r'}$ which also reach the eye. Therefore, in the case of such a source of light, the rays incident at different angles on the film are well adjusted by the eye and the field of view is large. Due to this reason, to observe interference phenomenon in this film, a broad source of light is required. With a broad source of light, rays of light are incident at different angles and the reflected parallel beams reach the eyes or the microscope objective. Each such ray of light has its origin at a different point on the source.

Types of Fringes Formed in Thin Film

Path difference between interfering waves= $\Delta = 2d \cos \theta$

Fringes of equal inclination or Haidinger fringes:

- These type of fringes are formed when a thin film of uniform thickness is illuminated by monochromatic light .
- In this case μt and λ are constant.
- So the fringes are locus of those points where angle of inclination is same.
- Hence they are also known as fringes of equal inclination.
- These fringes are circular in shape.
- These fringes are non localized.
- Example: Circular fringes in Michelson Interferometer.

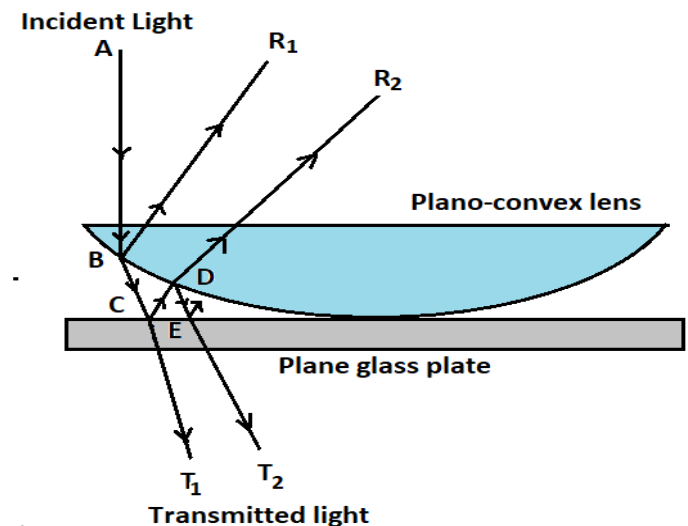
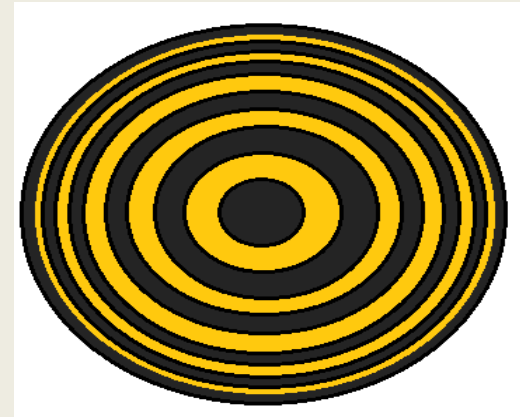


Types of Fringes Formed in Thin Film

$$\text{Path difference between interfering waves} = \Delta = 2\mu t \cos r$$

Fringes of equal width or Fizeau fringes:

- These type of fringes are formed when a thin film of variable thickness is illuminated by monochromatic light.
- In this case angle of incidence is kept constant and λ is constant.
- So, fringes originate for different values of μt and n i.e. these are locus of those points where $\mu t = \text{constant}$.
- Hence they are also known as fringes of equal thickness.
- The shape of fringes depends on the geometry of film.
- These fringes are localized.
- Example: Fringes obtained in Newton's ring experiment, straight line fringes and hyperbolic fringes in Michelson Interferometer.



Interference Due to a Thin Wedge – Shaped Film

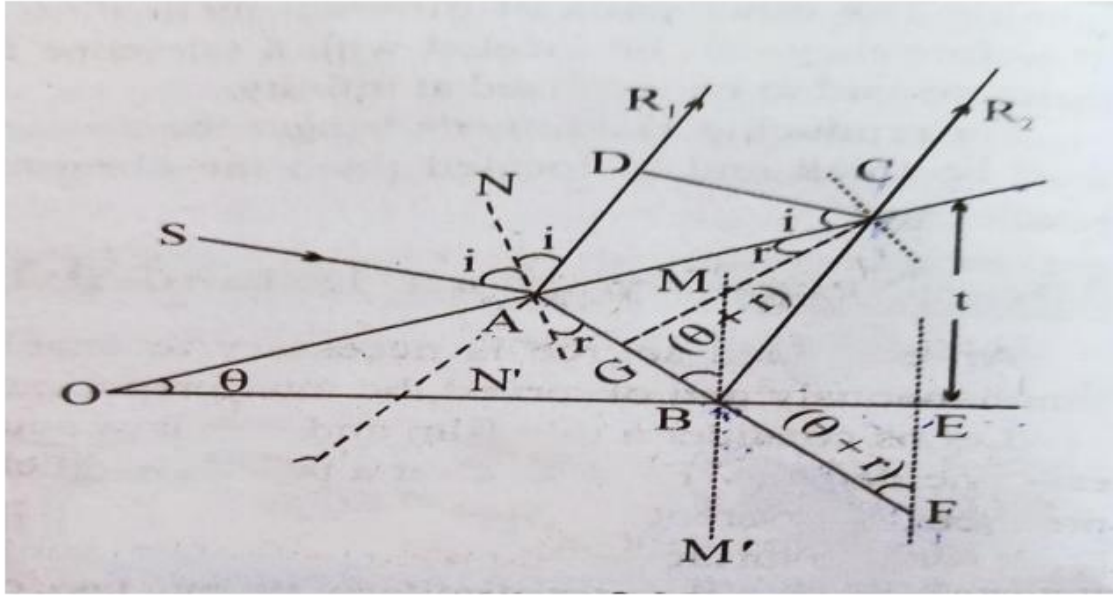


Fig – a



Let us consider a thin wedge shaped film of refractive index μ , bounded by two plane surfaces inclined at an angle θ (fig. a). Let a parallel beam of monochromatic light illuminate the film then the interference occurs between the incident ray reflected at the upper and lower surfaces of the film so that equidistant alternate dark and bright fringes become visible.

The interfering rays in reflected light are AR_1 and CR_2 , both originating from the same incident ray SA . To evaluate the path difference between these two rays the perpendiculars CD and CG are drawn from C on AR_1 and AB . If θ is an angle of wedge – shaped film is very small AR_1 and CR_2 will be almost parallel and after the perpendicular CD the paths will be equal. The perpendiculars CD and CG are drawn from C on AR_1 and AB to evaluate the path difference between these two rays

The path difference between AR1 and CR2 will be

$$\Delta 1 = 2 \mu t \cos(\theta + r)$$

Where θ is an angle of wedge – shaped film

The ray is reflected from a denser medium a phase change of π or a path difference is $\lambda/2$. Hence between the interfering rays AR1 and CR2 is

$$\Delta = \Delta 1 - \lambda/2$$

$$\Delta = 2\mu t \cos(\theta + r) - \lambda/2$$

While r and θ are very small. So the path difference primarily depends on the thickness of the film t .

If the incident ray of light is perpendicular to the wedge-shaped film $\angle i = \angle r = 0$, Then the effective path difference

$$\Delta = 2\mu t \cos\theta - \lambda/2$$

The path difference for constructive interference and Destructive interference are

Constructive interference	Destructive interference
The path difference is $\Delta = n\lambda$ ($n = 0, 1, 2, 3, \dots$) Or $2\mu t \cos\theta - \lambda/2 = n\lambda$ $2\mu t \cos\theta = (2n + 1) \lambda/2$	The path difference is $\Delta = (2n - 1) \lambda / 2$ $2\mu t \cos\theta - \lambda/2 = (2n - 1) \lambda / 2$ ($n = 1, 2, \dots$) $2\mu t \cos\theta = n\lambda$
When this condition is satisfied, the film will appear bright in the refracted light.	Under this condition, the film will appear dark in the reflected light.

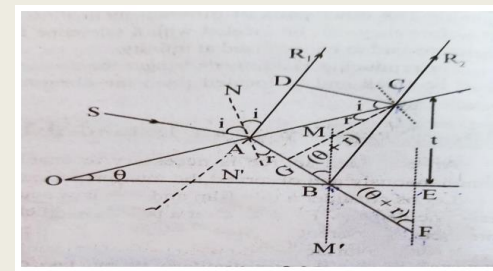
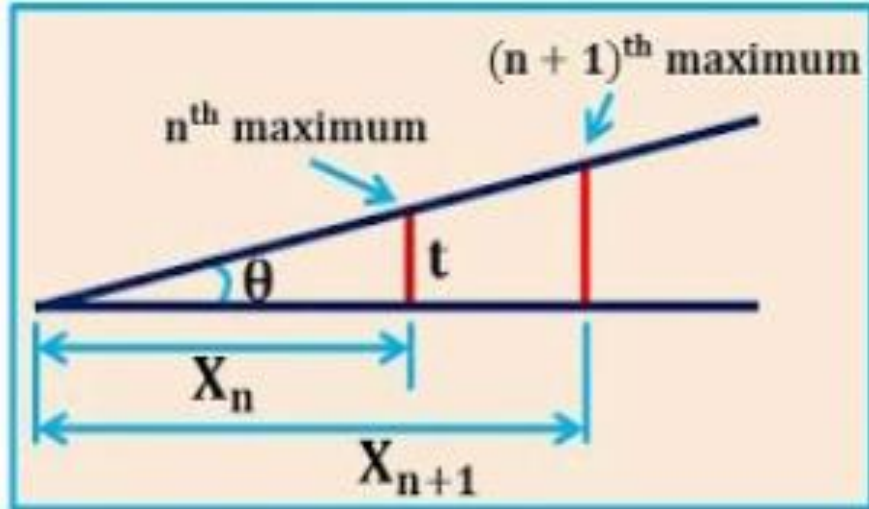


Fig – a

The thickness of film in the wedge-shaped film increases from zero as the distance from the edge increases. Therefore the path difference between the interfering rays increases from zero with the increasing of the thickness of film. Thus condition (7) and (8) are satisfied Alternately as the thickness of the film increases. Hence dark and bright fringes are alternately in the wedge-shaped film.

Determination of Fringe width for wedge-shaped

Relation for fringe width



For n^{th} maximum, we have -

$$2 \mu t \cos(r + \theta) = (2n - 1) \frac{\lambda}{2} \quad n = 1, 2, 3 \dots$$

For normal incidence, $r = 0$

$$\therefore 2 \mu t \cos \theta = (2n - 1) \frac{\lambda}{2} \quad (1)$$



Let n^{th} maximum is obtained at distance X_n from the edge

From figure, $\tan \theta = \frac{t}{X_n} \Rightarrow t = X_n \tan \theta$

(1) becomes

$$\therefore 2 \mu X_n \tan \theta \cos \theta = (2n - 1) \frac{\lambda}{2}$$

$$\therefore 2 \mu X_n \sin \theta = (2n - 1) \frac{\lambda}{2} \quad (2)$$

Similarly, for $(n + 1)^{\text{th}}$ maximum,

$$\therefore 2 \mu X_{n+1} \sin \theta = (2(n + 1) - 1) \frac{\lambda}{2} \quad (3)$$

Subtract (2) from (3)

$$X_{n+1} - X_n = \frac{\lambda}{2 \mu \sin \theta} = \text{fringewidth } (\beta)$$

if wedge angle is very small $\sin \theta \approx \theta$, $\beta = \lambda / 2 \mu \sin \theta \approx \lambda / 2 \mu \theta$,
So the **Fringe width** $\beta = \lambda / 2 \mu \theta$

Application of Interference Due to a Thin Wedge – Shaped Film :-

Testing the plane of surfaces

For testing the planeness of surface of a glass plate it is placed on an optically plane surface in such a way that a wedge shaped air film of very small wedge angle is formed between these surfaces. Now it is illuminated with monochromatic light. The fringes so produces, are observed in the study of microscope and if the fringes are of equal width and straight, it means that the testing surface is plane otherwise it is not plane.

Summary

- In wave optics light is considered to be propagating as a wave.
- The waves emitted by two sources having same frequency and constant phase difference are said to be coherent and this phenomenon is known as coherence.
- Coherent sources can be produced either by division of wave front or division of amplitude.
- Redistribution of intensity in space due to superposition of two or more waves of nearly same amplitude, same frequency and having constant phase difference is known as interference of light.
- An electromagnetic wave undergoes a phase change of 180° upon reflection from an optically denser medium.

Summary

➤ Thin film interference in reflected light:

Effective path difference between partially reflected rays

$$\Delta_{eff} = 2\mu t \cos r - \frac{\lambda}{2}$$

Intensity will be maximum when

$$2\mu t \cos r = \frac{(2n+1)\lambda}{2}$$

Intensity will be minimum when

$$2\mu t \cos r = n\lambda$$

Where $n=0,1,2,3,\dots$

➤ Thin film interference in transmitted light:

Effective path difference between partially reflected rays

$$\Delta_{eff} = 2\mu t \cos r$$

Intensity will be maximum when

$$2\mu t \cos r = n\lambda$$

Where $n=1,2,3,\dots$

Intensity will be minimum when

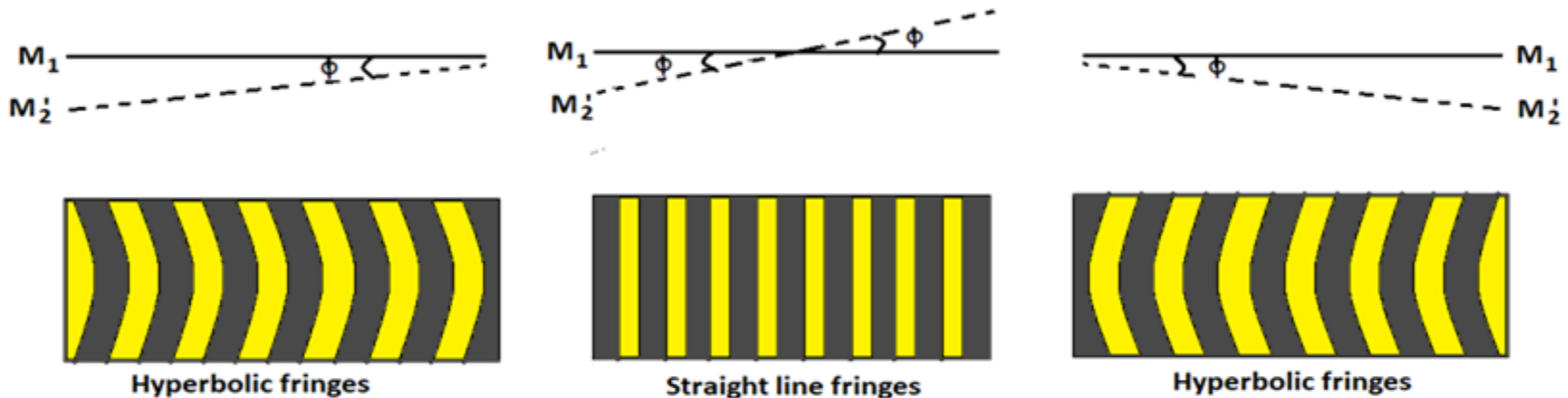
$$2\mu t \cos r = \frac{(2n-1)\lambda}{2}$$

Where $n=1,2,3,\dots$

➤ The interference fringe pattern produced due to superposition of partially reflected rays is complementary of fringe pattern produced due to superposition of partially transmitted rays.

ONLY for GK: Types of Fringes Formed in Thin Film

Fringes of equal width or Fizeau fringes:



Shape of localized fringes in Michelson's Interferometer

**Thank
You**

